



Analyse II (proposé par Mohamed Hassen)

$$= \frac{x+1}{(x^2+2x)^3}$$

$$u = x^2+2x$$

$$du = 2x+2 \cdot dx$$

$$\Rightarrow du = 2(x+1) \cdot dx$$

$$\Rightarrow \int \frac{1}{2u^3} \cdot du$$

$$\Rightarrow \frac{1}{2} \cdot \int \frac{1}{u^3} \cdot du \Rightarrow \frac{1}{2} \left[-\frac{1}{2u^2} \right] \Rightarrow \frac{1}{2} \left[\frac{-1}{2(x^2+2x)^2} \right]$$

$$2. f(x) = \frac{x}{x^2-1}$$

Méthode 1:

$$\frac{x}{x^2-1} = \frac{A}{x+1} + \frac{B}{x-1} \Rightarrow \frac{A(1+x) + B(1-x)}{(1-x)(1+x)}$$

$$\Rightarrow \frac{A+A x+B-B x}{x^2-1} \Rightarrow \frac{(A+B) + (A-B)x}{x^2-1} = \frac{1}{x^2-1}$$

$$\Rightarrow (A+B) + (A-B)x = 1 = 1 + x \cdot 0$$

$$\Rightarrow \begin{cases} A+B=1 \\ A-B=0 \end{cases} \Rightarrow \begin{cases} A+A=1 \Rightarrow A=1/2 \\ A=B \end{cases}$$

$$\Rightarrow \frac{1}{2} \left(\frac{1}{x-1} \right) + \frac{1}{2} \left(\frac{1}{x+1} \right) \Rightarrow \frac{1}{2} \ln \left| \frac{1+x}{x-1} \right| + C$$

Méthode 2:

$$u = x^2-1$$

$$\begin{cases} du = 2x \cdot dx \Rightarrow dx = \frac{1}{2u} \end{cases} \Rightarrow \int \frac{1}{2u} \cdot du \Rightarrow \frac{1}{2} \int \frac{1}{u} \cdot du$$

$$\Rightarrow \frac{1}{2} [\ln(u)] + C$$

Exo 2:

$$f(x) = (\sin^2 x - 3 \sin x + 3) \cos(x) \quad ; \quad F(3\pi/2) = 0$$

$$\Rightarrow F(x) = \int_a^x f(t) dt \Rightarrow \int_a^x (\sin^2 t - 3 \sin t + 3) \cos t$$

$$\Rightarrow \int_a^x \sin^2 t \cos t - 3 \sin t \cos t + 3 \cos t \Rightarrow \left[\frac{1}{3} \sin^3 t - \frac{3}{2} \sin^2 t + 3 \sin t \right]$$

$$\Rightarrow \frac{1}{3} \sin^3 x - \frac{3}{2} \sin^2 x + 3 \sin x + C \Rightarrow -\frac{1}{3} - \frac{3}{2} - 3 + C = 0$$

$$\Rightarrow \boxed{C = 29/6}$$

Exo38



$$(x^2 + 2x + 1) + 1 \Rightarrow x^3 + 2x^2 + x + 3x^2 + 6x + 3 + 1 + 5x^2 + 7x + 4$$

$$F(x) = \int_a^x \frac{t^3 + 5t^2 + 7t + 4}{t^2 + 2t} dt \Rightarrow \int_a^x \frac{(t+3)(t^2 + 2t + 1) + 1}{t^2 + 2t + 1}$$

$$\Rightarrow \int_a^x t + 3 + \frac{1}{(t+1)^2} dt \Rightarrow \left[\frac{t^2}{2} + 3t - \frac{1}{t+1} \right]_a^x$$

Exo40

$$f(x) = \begin{cases} -1 & \text{si } x=0 \\ 1 & \text{si } 0 < x < 1 \\ 3 & \text{si } x=1 \\ -2 & \text{si } 1 < x \leq 2 \\ 4 & \text{si } 2 \leq x \leq 4 \end{cases}$$

$$\begin{aligned} &\Rightarrow \int_0^4 f(t) dt \Rightarrow \int_0^1 f(t) dt + \int_1^2 f(t) dt + \int_2^4 f(t) dt \\ &+ \int_2^4 4 dt \Rightarrow [t]_0^1 + [-2t]_1^2 + [4t]_2^4 = 1 - 4 + 2 + 16 - 8 = 7 \end{aligned}$$

Exo50

$$1. \int x^2 \ln x \Rightarrow \begin{cases} u = \ln(x) \rightarrow du = \frac{1}{x} \\ dv' = x^2 \rightarrow v = \frac{x^3}{3} \end{cases} \Rightarrow \left[\ln(x) \cdot \frac{x^3}{3} \right] - \int \frac{x^3}{3} \cdot \frac{1}{x} dx$$

$$\Rightarrow \left[\ln(x) \cdot \frac{x^3}{3} \right] - \frac{1}{3} \left[\frac{x^3}{3} \right]$$

$$2. \int x \arctan x \Rightarrow \begin{cases} u = \arctan(x) \rightarrow u' = \frac{1}{1+x^2} \\ v' = x \rightarrow v = \frac{x^2}{2} \end{cases}$$

$$\Rightarrow \int x \arctan x = \left[\frac{x^2}{2} \arctan x \right] - \frac{1}{2} \int \frac{x^2 + 1 - 1}{1 + x^2} \cdot dx$$

$$\Rightarrow \left[\frac{x^2}{2} \arctan x \right] - \frac{1}{2} \left(\int \frac{x^2 + 1}{x^2 + 1} \cdot dx + \int \frac{-1}{1 + x^2} \cdot dx \right)$$

$$\Rightarrow \left[\frac{x^2}{2} \arctan x \right] - \frac{1}{2} \left[x - \arctan(x) \right]$$

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(Suite Exo 5)



$$x \Rightarrow \begin{array}{l} u = \ln x \rightarrow u' = \frac{1}{x} \\ v' = 1 \rightarrow v = x \end{array} \Rightarrow [x \ln x] - \int 1 \cdot dx$$

$$\Rightarrow [x \ln x] - [x] \leftarrow$$

$$4) \int (\ln x)^2 \cdot dx \Rightarrow \begin{array}{l} u = \ln(x)^2 \rightarrow u' = 2 \ln(x) \cdot \frac{1}{x} \\ v' = 1 \rightarrow v = x \end{array}$$

$$\Rightarrow [x \ln(x)^2] - \left[2 \int \ln x \right]$$

$$\Rightarrow [x \ln(x)^2] - 2[x \ln(x) - x]$$

$$5) \int (\cos x) \cdot e^x \Rightarrow [\sin x \cdot e^x] - \int e^x \cdot \sin x$$
$$[e^x \cdot \cos x] - \int -\cos x \cdot e^x$$

$$\Rightarrow \int \cos(x) \cdot e^x = [e^x \cdot \sin(x)] + [e^x \cdot \cos(x)] - \int \cos(x) \cdot e^x$$

$$2 \int \cos(x) \cdot e^x = e^x (\sin(x) + \cos(x))$$

$$\Rightarrow \int \cos(x) \cdot e^x = \frac{1}{2} e^x (\sin(x) + \cos(x))$$

Exo 6:

$$1. \int_0^{\pi/2} x \sin(x) \cdot dx \Rightarrow [-x \cos x]_0^{\pi/2} - \int_0^{\pi/2} -\cos x \cdot dx$$

$$\Rightarrow [-x \cos x]_0^{\pi/2} + [\sin x]_0^{\pi/2}$$

$$2. \int_0^1 \frac{e^x}{\sqrt{e^x + 1}} \cdot dx \Rightarrow \begin{cases} u = e^x + 1 \\ du = e^x \cdot dx \end{cases} \Rightarrow \int_0^1 \frac{1}{\sqrt{u}} \cdot du$$

$$\Rightarrow [2u^{1/2}]_0^1 \Rightarrow$$

$$3. \int_0^1 \frac{1}{(1+x^2)^2} \cdot dx \Rightarrow \begin{cases} x = \tan(u) = \frac{\sin(u)}{\cos(u)} \\ dx = \frac{1}{\cos^2 u} \cdot du \end{cases}$$

$$\Rightarrow \int_0^1 \frac{1}{\left(1 + \frac{\sin^2 u}{\cos^2 u}\right)^2} \cdot \frac{1}{\cos^2 u} \cdot du \Rightarrow \int_0^1 \frac{1}{\left(\frac{1}{\cos^2 u}\right)^2} \cdot \frac{1}{\cos^2 u} \cdot du$$



$$\int_0^1 \frac{\cos^4 u}{\cos^2 u} \cdot du \Rightarrow \int_0^1 \cos^2 u \cdot du \Rightarrow \left| \cos^2 u = \frac{1}{2} (1 + \cos(2u)) \right|$$

$$\int_0^1 (1 + \cos(2u)) \cdot du = \frac{1}{2} \left(\int_0^1 1 \cdot du + \int_0^1 \cos(2u) \cdot du \right)$$

$$\Rightarrow \frac{1}{2} \left(u + \frac{1}{2} \sin(2u) \right)$$

$$\sin(2u) = 2 \sin u \cdot \cos u$$

$$\Rightarrow \frac{1}{2} \left(u + \frac{1}{2} (2 \sin u \cdot \cos u) \right)$$

$$4) \int_0^1 \frac{3x+1}{(1+x)^2} \cdot dx \Rightarrow \int_0^1 \frac{3(x+1-1)+1}{(1+x)^2} \cdot dx$$

$$\Rightarrow \int_0^1 \frac{3(x+1) - 3 + 1}{(1+x)^2} \cdot dx \Rightarrow \int_0^1 \frac{3}{1+x} - \frac{2}{(1+x)^2} \cdot dx$$

$$\left[3 \ln(1+x) + \frac{2}{1+x} \right]_0^1$$

Exo 7:

$$1) \frac{u^2-1}{2u-1} = au + b + \frac{c}{2u-1} \Rightarrow \frac{(au+b)(2u-1) + c}{2u-1}$$

$$\Rightarrow \begin{cases} 2a=1 \\ 2b-a=0 \\ -b+c=0 \end{cases} \Rightarrow \begin{cases} a=1/2 \\ b=1/4 \\ c=3/4 \end{cases}$$

$$2) \int_{-1}^0 \frac{u^2-1}{2u-1} = \int_{-1}^0 \left(\frac{1}{2}u + \frac{1}{4} + \frac{3/4}{2u-1} \right) \cdot du \Rightarrow \left[\frac{1}{4}u^2 + \frac{1}{4}u - \frac{3}{8} \ln|2u-1| \right]_{-1}^0$$

$$3) \int_{-\pi/6}^0 \frac{\cos^3 x}{1-2\sin x} \cdot dx \Rightarrow \int_{-\pi/6}^0 \frac{\cos^2 x \cdot \cos x}{1-2\sin x} \cdot dx \Rightarrow \int_{-\pi/6}^0 \frac{1-\sin^2 x}{1-2\sin x} \cdot \cos x$$

$$\Rightarrow u = \sin x \Rightarrow du = \cos x \cdot dx \Rightarrow -x=0 \Rightarrow u = \sin(0) = 0$$

$$\Rightarrow x = -\pi/6 \Rightarrow u = \sin(-\pi/6) = -1/2$$

$$\Rightarrow \int_{-1/2}^0 \frac{1-u^2}{1-2u} \cdot du \Rightarrow \int_{-1/2}^0 \frac{u^2-1}{2u-1} \cdot du \Rightarrow \left[\frac{1}{4}u^2 + \frac{1}{4}u - \frac{3}{8} \ln|2u-1| \right]_{-1/2}^0$$

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$$F(t) = \int_{\varphi(t)}^{\psi(t)} f(x) \cdot dx \text{ alors } F'(t) = \psi'(t) \cdot f(\psi(t)) - \varphi'(t) \cdot f(\varphi(t))$$

$$F(t) = \int_{-\tan t}^{8ht} \frac{1}{\sqrt{1+x^2}} \cdot dx \Rightarrow F'(t) = (8ht)' \cdot \frac{1}{\sqrt{1+8h^2t^2}} - \frac{(-\tan t)'}{\sqrt{1+\tan^2 t}}$$

$$= \frac{dh \cdot t}{\sqrt{1+8h^2t^2}} + \frac{(1+\tan^2 t)}{\sqrt{1+\tan^2 t}}$$

Exo 9

$$1) \int_{-1}^1 x|x| \cdot dx \Rightarrow \int_{-1}^0 x|x| \cdot dx + \int_0^1 x|x| \cdot dx = \int_{-1}^0 -x^2 \cdot dx + \int_0^1 x^2 \cdot dx$$

$$\Rightarrow \left[-\frac{x^3}{3} \right]_{-1}^0 + \left[\frac{x^3}{3} \right]_0^1 = 0$$

$$2) \int_{-1}^2 x|x| \cdot dx \Rightarrow \int_{-1}^1 x|x| \cdot dx + \int_1^2 x|x| \cdot dx \Rightarrow \int_1^2 x^2 \cdot dx = \left[\frac{x^3}{3} \right]_1^2$$

Exo 10

$$1) \int_0^1 \frac{1}{\sqrt{1-x^2}} \cdot dx = \left[\arcsin x \right]_0^1$$

Démonstration:

$$x = \sin(u) \rightarrow dx = \cos(u) \cdot du \quad \left| \begin{array}{l} \sin^2 u + \cos^2 u = 1 \\ \Rightarrow \cos^2 u = 1 - \sin^2 u \end{array} \right.$$

$$\Rightarrow \int_0^1 \frac{1}{\sqrt{1-\sin^2 u}} \cdot \cos(u) \cdot du \Rightarrow \int_0^1 \frac{\cos(u)}{\sqrt{\cos^2 u}} \cdot du$$

$$\Rightarrow \int_0^1 1 \cdot du \Rightarrow [u]_0^1 \quad \left| \text{si } x = \sin u \text{ alors } u = \sin^{-1} x \right.$$

$$2) \int_1^2 \frac{1}{\sqrt{x^2-1}} \cdot dx \Rightarrow x = \frac{1}{\cos(u)} \Rightarrow dx = \frac{\sin(u)}{\cos^2 u} \cdot du$$

$$\Rightarrow \int_1^2 \frac{1}{\sqrt{\frac{1}{\cos^2 u} - 1}} \cdot \frac{\sin(u)}{\cos^2 u} \cdot du \Rightarrow \int_1^2 \frac{1}{\frac{\sin u}{\cos u}} \cdot \frac{\sin u}{\cos^2 u} \cdot du \Rightarrow \int_1^2 \frac{1}{\cos u} \cdot du$$

$$\Rightarrow \int_1^2 \frac{1}{\cos(u)} \cdot du \Rightarrow \left[\ln |\tan(u) + \frac{1}{\cos(u)}| \right]_1^2$$



$$3) \int_0^1 \frac{x}{1+x^4} \cdot dx \Rightarrow \frac{1}{2} \int_0^1 \frac{2x}{1+x^4} \cdot dx \quad \left| \begin{array}{l} u = x^2 \\ du = 2x \cdot dx \end{array} \right.$$

$$\Rightarrow \frac{1}{2} \int_0^1 \frac{1}{1+u^2} \cdot du \Rightarrow \frac{1}{2} \left[\arctan(u) \right]_0^1$$

$$4) \int_e^{e^2} \frac{1}{x(\ln(x))^2} \cdot dx \Rightarrow \begin{array}{l} u = \ln(x) \rightarrow e^u = x \\ du = \frac{1}{x} \cdot dx = \frac{1}{e^u} \cdot dx \end{array} \Rightarrow e^u \cdot du = dx$$

$$\Rightarrow \int_e^{e^2} \frac{1}{e^u \cdot u^2} \cdot e^u \cdot du \Rightarrow \int_e^{e^2} \frac{1}{u^2} \cdot du \Rightarrow \left[\frac{-1}{\ln x} \right]_e^{e^2}$$